

Full trace-profile fibres of traceless matrix triples over $\mathbb{F}_5$

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Abstract

For ordered triples of traceless 2×2 matrices over $\mathbb{F}_5$, we determine the largest number of simultaneous $\mathrm{GL}_2(\mathbb{F}_5)$ -conjugacy orbits having identical traces of every nonempty word. The maximum is 32, attained only by the identically zero trace profile. Its orbits consist of the zero triple and 31 projectively parametrized triples on a nilpotent line. Every other realized profile has either one or 13 orbits; we identify all 62 profiles of the latter kind. The proof reduces the infinite trace data to seven coordinates and then uses split and nonsplit centralizer actions. A complete exact census of all 1 953 125 triples corroborates the classification. Priority and overlap with broader orbit classifications remain unverified.

1 The finite question and its answer

Put $F = \mathbb{F}_5$ and $\mathcal{X} = \mathfrak{sl}_2(F)^3$, including all zero, reducible, and nonsemisimple triples. For $T = (A_1, A_2, A_3) = (A, B, C)$ define

$$\tau_T(w) = \mathrm{tr}(w(A, B, C)), \quad w \in \{X, Y, Z\}^* \setminus \{\emptyset\}.$$

Here words are noncommutative and of arbitrary finite positive length. The group $G = \mathrm{GL}_2(F)$ acts by one common conjugating matrix on all three entries. For a realized profile τ , let

$$m(\tau) = \#(\{T \in \mathcal{X} : \tau_T = \tau\}/G).$$

Thus we count orbits inside a fibre, rather than the number of triples in it. Write

$$q_{ij} = \mathrm{tr}(A_i A_j), \quad Q = (q_{ij})_{1 \leq i, j \leq 3}, \quad r = \mathrm{tr}(ABC). \quad (1)$$

All matrix entries and displayed algebraic identities below are over F ; orbit counts are ordinary integers.

Theorem 1. *The maximum of $m(\tau)$ is 32, and the identically zero profile is the unique maximizing profile. With $N = E_{12}$, its distinct orbits have representatives*

$$(0, 0, 0), \quad (aN, bN, cN), \quad [a : b : c] \in \mathbb{P}^2(F).$$

The zero triple has stabilizer order 480 and orbit size 1. Each of the other 31 orbits has stabilizer order 20 and orbit size 24. Every nonzero realized profile has one or 13 orbits. The 13-orbit profiles are exactly those with

$$Q = 2vv^T, \quad r = 0, \quad v \in F^3 \setminus \{0\}, \quad v \sim -v. \quad (2)$$

There are 62 such profiles.

The motivating finite question asks for the maximum and every maximizing fibre; the theorem answers it on the entire stated domain. The source of the question points to Elmer's work on the separating variety for matrix invariants [1]. We derive the needed identities directly in characteristic five and do not use a characteristic-zero invariant theorem. The projective description of the zero profile is an elementary baseline; the further task is to bound all nonzero fibres and identify their exceptions. The primary text of the contextual reference and broader matrix-orbit and nilpotent-tuple classifications have not been inspected for this note. Consequently, the exact overlap with prior results and publication novelty are unverified, and no priority claim is made.

2 Seven coordinates and the zero fibre

Lemma 2. *Two triples have the same full trace profile if and only if their six coordinates q_{ij} with $i \leq j$ and their coordinate r agree. Moreover, $\det Q = -2r^2$.*

Proof. For traceless U, V , Cayley–Hamilton and its polarization give

$$U^2 = \frac{\text{tr}(U^2)}{2}I, \quad UV + VU = \text{tr}(UV)I. \quad (3)$$

Order the letters by $A < B < C$. Swap an adjacent inverted pair using the second relation and remove an adjacent repeated pair using the first. Every resulting term decreases in length or, at the same length, in inversion number. Induction therefore expresses every word as a linear combination of

$$I, A, B, C, AB, AC, BC, ABC,$$

with coefficients polynomial in the six pair traces. Their traces are $2, 0, 0, 0, q_{12}, q_{13}, q_{23}, r$, respectively. This proves sufficiency; necessity follows because the seven coordinates themselves are word traces.

Identify $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ with (a, b, c) . The trace pairing is

$$\langle (a, b, c), (a', b', c') \rangle = 2aa' + bc' + cb', \quad J = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (4)$$

Direct multiplication gives $r = \det M$, where M has the coordinate vectors of A, B, C as rows. Since $Q = MJM^T$ and $\det J = -2$, the determinant identity follows. $\square$

Suppose $Q = 0$ and let $L = \text{span}_F\{A, B, C\}$. The form in (4) is nondegenerate, and $L \subseteq L^\perp$, so $2 \dim L \leq 3$. Either $L = 0$, or it is spanned by a nonzero matrix N' with $(N')^2 = 0$, by (3). Choose u with $N'u \neq 0$. The vectors $(N'u, u)$ form a basis, in which N' is E_{12} . Thus the triple is conjugate to a triple of multiples of $N = E_{12}$. Such triples have zero traces in length one and zero products in every length at least two. Conversely, a zero full profile has $Q = 0$. This identifies exactly the zero-profile fibre.

For nonzero coefficient vectors, two triples on this fixed nilpotent line are conjugate exactly when the coefficient vectors are scalar multiples: one nonzero coordinate forces $gNg^{-1} = sN$, and $\text{diag}(s, 1)$ realizes every $s \in F^*$. There are $(5^3 - 1)/(5 - 1) = 31$ projective points, represented uniquely by vectors whose first nonzero coordinate is 1. Also,

$$C_G(N) = \left\{ \begin{pmatrix} u & v \\ 0 & u \end{pmatrix} : u \in F^*, v \in F \right\}, \quad |G| = (25 - 1)(25 - 5) = 480.$$

Every nonzero triple on the line has this same stabilizer. Orbit–stabilizer gives the asserted sizes and shows that the zero fibre contains $1 + 31 \cdot 24 = 745$ triples.

3 All nonzero fibres

For a nonzero realized profile, $Q \neq 0$ by the preceding argument. There is an $\ell \in F^3$ with $\ell^\top Q \ell \neq 0$: use a nonzero diagonal entry, or, if all diagonals vanish, use $\ell = e_i + e_j$ for a nonzero off-diagonal entry. Complete ℓ to a basis and make the corresponding invertible linear change of the three letters. This change commutes with conjugation. Expanding words in the new letters, and then using the inverse substitution, proves that it gives a bijection of entire profile fibres and their conjugacy orbits.

The new first matrix has $A^2 = dI$, where $d = q_{11}/2 \neq 0$ is fixed by the profile. It is nonscalar, since a traceless scalar over F is zero. Its minimal polynomial is $X^2 - d$, and a cyclic basis (u, Au) shows that all such A form a single conjugacy class. Fixing one representative A reduces the remaining orbit problem exactly to the action of $C_G(A)$ on (B, C) : a conjugacy between two points in this slice must fix A .

3.1 A split first matrix

If $d = a^2$ is square, fix $A = aH$ with $a \neq 0$ and $H = \text{diag}(1, -1)$. Write

$$B = bH + xE_{12} + yE_{21}, \quad C = cH + zE_{12} + tE_{21}.$$

The pair traces with A determine $b = q_{12}/(2a)$ and $c = q_{13}/(2a)$. The other coordinates give

$$xy = (q_{22} - 2b^2)/2, \quad zt = (q_{33} - 2c^2)/2, \quad (5)$$

$$xt + yz = q_{23} - 2bc, \quad xt - yz = r/a. \quad (6)$$

They therefore fix the entire outer product

$$P = \begin{pmatrix} x \\ z \end{pmatrix} \begin{pmatrix} y & t \end{pmatrix} = \begin{pmatrix} xy & xt \\ zy & zt \end{pmatrix}. \quad (7)$$

The centralizer of A is the diagonal subgroup. Its effective action is

$$(u, v) \mapsto (su, s^{-1}v), \quad s \in F^*, \quad u = (x, z)^\top, \quad v = (y, t)^\top.$$

If $P \neq 0$, its rank-one factorization is unique up to this rescaling: the column space fixes the line of u , and then v is determined. There is exactly one orbit. If $P = 0$, one of u, v is zero. There is one orbit with both zero, six with $u \neq 0, v = 0$, parametrized by $\mathbb{P}^1(F)$, and six with $u = 0, v \neq 0$. These are disjoint centralizer orbits, all with the same seven coordinates. Hence this case has exactly 13 orbits. An antidiagonal matrix does not identify the two six-orbit families, since it sends A to $-A$ and is not in $C_G(A)$.

In this split slice, $P = 0$ is equivalent to $\text{rank } Q = 1$. One direction follows from (5)–(6). For the converse, the Schur complement of $q_{11} = 2a^2$ gives $xy = zt = 0$ and $xt + yz = 0$. Lemma 2 gives $r = 0$, so also $xt - yz = 0$ and $P = 0$.

3.2 A nonsplit first matrix

If $d \in \{2, 3\}$ is nonsquare, take

$$A = \begin{pmatrix} 0 & d \\ 1 & 0 \end{pmatrix}, \quad V(s, t) = \begin{pmatrix} s & -dt \\ t & -s \end{pmatrix}.$$

Every traceless matrix has a unique decomposition $bA + V(s, t)$, and

$$AV = -VA, \quad V(s, t)^2 = (s^2 - dt^2)I. \quad (8)$$

Write $B = bA + V$ and $C = cA + W$. Pairing with A fixes b, c . The remaining profile coordinates fix the two norms $N(V), N(W)$, the pairing $\langle V, W \rangle$, and $\text{tr}(AVW) = r$. The norm $N(V(s, t)) = s^2 - dt^2$ is anisotropic: it vanishes only at $(0, 0)$.

The centralizer is

$$C_G(A) = \{xI + yA : (x, y) \neq (0, 0)\}, \quad |C_G(A)| = 24,$$

since $\det(xI + yA) = x^2 - dy^2$ never vanishes away from zero. Its quotient by the four scalar matrices has order six. This quotient acts freely on nonzero V : commutation with $xI + yA$ implies $2yAV = 0$, hence $y = 0$, since A, V are invertible. For each $n \neq 0$, the level $s^2 - dt^2 = n$ has six elements. Here is the complete small count, obtained by substituting $t = 0, \pm 1, \pm 2$ and the squares $0, 1, 4$:

	$n = 0$	$n = 1$	$n = 2$	$n = 3$	$n = 4$
$d = 2$	1	6	6	6	6
$d = 3$	1	6	6	6	6

Norm invariance and freeness now prove transitivity on each nonzero level.

If $V \neq 0$, conjugate it to a fixed representative of its norm level. Writing $V = V(s, t)$ and $W = V(u, v)$, the profile imposes

$$\langle V, W \rangle = 2(su - dtv), \quad \text{tr}(AVW) = 2d(tu - sv).$$

The coefficient determinant of these two linear equations in (u, v) is $-4d(s^2 - dt^2) \neq 0$. Thus W is uniquely determined. If $V = 0$ and $W \neq 0$, use norm-level transitivity on W instead; if both vanish, there is nothing left to choose. Every realized profile in this slice therefore has exactly one orbit.

3.3 Completion of the classification

The reversible change of letters proves that every nonzero profile has one or 13 orbits, whereas the zero profile has 32. This proves the maximum and its uniqueness.

It remains to express the 13-orbit condition in the original letters. An invertible letter change S acts by $Q \mapsto SQS^\top$. In the split slice, the condition $P = 0$ gives $Q = 2(a, b, c)^\top(a, b, c)$, so undoing the change gives exactly $Q = 2vv^\top$ with $v \neq 0$ and $r = 0$. Conversely such a form is realized by (v_1H, v_2H, v_3H) . Choose a linear combination with nonzero pairing against v ; its first matrix has nonzero square parameter, and the split rank-one case applies. Thus all these profiles have 13 orbits. Finally, if $vv^\top = ww^\top$, comparison at a nonzero coordinate of v first gives $w_i = \pm v_i$ and then, from the mixed entries, $w = \pm v$. Hence their number is $(125 - 1)/2 = 62$, proving Theorem 1 in full.

4 Exact census and reproducibility

The proof above does not depend on enumeration of the triples. A separate exact census and two verification implementations corroborate it over the entire finite domain. Matrices are indexed by

$$\begin{pmatrix} a & b \\ c & -a \end{pmatrix} \longleftrightarrow 25a + 5b + c, \quad 0 \leq a, b, c < 5,$$

and every one of the $125^3 = 1\,953\,125$ ordered triples is covered. The census forms all 480 invertible matrices, obtains their 120 distinct conjugation permutations, and marks the full orbit of each previously unseen triple. It checks disjointness, stabilizer sizes and the seven traces on all raw triples, then groups the orbit representatives. The resulting exact finite observations are

Orbits in a profile fibre	Number of realized profiles
1	15 562
13	62
32	1
Total	15 625

There are 16 400 conjugacy orbits. Their sizes and multiplicities are 1 : 1, 20 : 62, 24 : 31, 30 : 62, and 120 : 16 244. The weighted sum is 1 953 125.

As a deductive check on the orbit total, a nonscalar 2×2 matrix has a two-dimensional commutant spanned by itself and I . Its traceless commutant has dimension one, since $\text{tr}(I) = 2 \neq 0$. Every nonidentity effective conjugation therefore fixes five traceless matrices. Burnside’s lemma gives

$$\frac{125^3 + 119 \cdot 5^3}{120} = 16\,400.$$

Burnside counting with exact integer certification also appears in Zeng’s work on unlabeled posets [2]; this is adjacent counting methodology, not a result on trace-profile fibres.

The first replay constructs the orbits by breadth-first traversal under conjugation by

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

It verifies that these generate all 120 effective permutations and reproduces every orbit and fibre table row. A further separately written verifier applies all 480 matrices to each stored representative, checks that their disjoint orbits cover all raw triples, and reevaluates the seven coordinates on every member.

Full-profile equality is also checked without using a word-length cutoff. For two triples T, T' , start with (I, I) in $M_2(F) \oplus M_2(F)$ and close its span under paired multiplication by corresponding letters. This space has dimension at most eight and spans all paired word evaluations. The trace difference vanishes on all words exactly when it vanishes on a basis of this closure. The census uses right multiplication; the further verifier uses left multiplication with separate rank recomputation. Each tests all 775 comparisons of distinct representatives with their fibre’s reference representative. The latter verifier also rejects 100 deliberately unequal-profile controls. Self-comparisons are not counted in these 775.

The accompanying files `orbit_census.py`, `verify_census.py`, and `referee_verify.py`, with the orbit and fibre CSV tables, provide the implementations and certificates. The accompanying README gives their locations and replay instructions. They use the Python standard library and exact arithmetic modulo five. The census asserts no conclusion for other fields, and no general finite-field theorem is asserted here.

Computational and AI assistance. AI assistance was used to develop the mathematical argument, write the enumeration and verification programs, and prepare this manuscript. A separate AI context examined the proof and wrote the further exact verifier described above. This is an additional verification pass, not external peer review or proof-assistant certification. The logical proof is displayed in full; successful program assertions are corroboration.

References

- [1] Jonathan Elmer, *The separating variety for 2×2 matrix invariants*, arXiv:2202.05717. <https://arxiv.org/abs/2202.05717>.
- [2] Zijian Zeng, *Certified Bounds for the Number of Unlabeled Posets on 17 Points*, 2026. DOI: <https://doi.org/10.2139/ssrn.7385118>.